



Agentic AI

Make AI “Work”, not only Answer

Whoami



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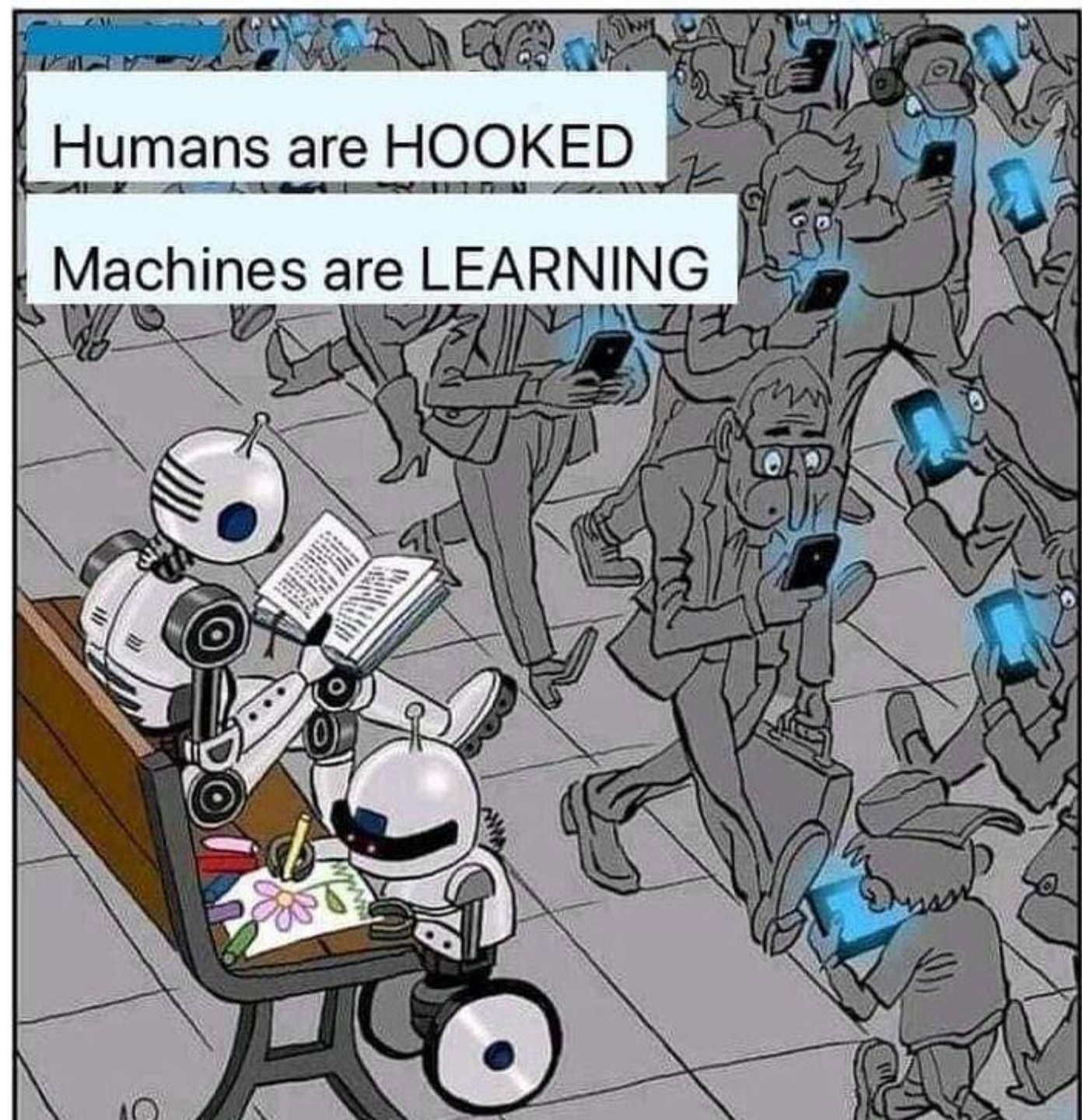
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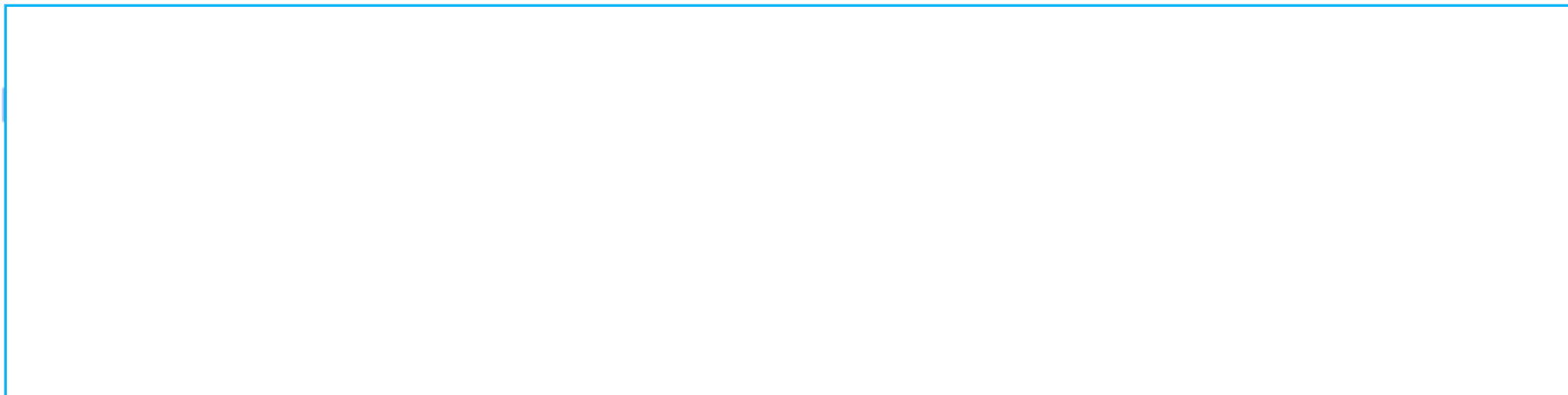
See the portofolio on <https://me.daudaldi.my.id>

Agenda

1. Revisit Previous Materi
2. Agentic AI 101
3. DIY Portofolio



Timeline Sejarah AI



Agentic AI

2023 - Now
AI otonom yang
berinteraksi

Masalah Sebelum Agentic Era



**Jarvis, tolong anunya
Sedikit digitukan**



Contoh Penggunaan

The screenshot displays a Kali Linux desktop environment with a terminal window and a web browser showing the HTB Labs interface.

Terminal Window:

```
kali@barengsinau: ~/lab/ailab/HTB_Challenge
(kali@barengsinau)~[~/lab/ailab/HTB_Challenge]
$ ls easy_rev_simple_encryptor
rev_simpleencryptor
(kali@barengsinau)~[~/lab/ailab/HTB_Challenge]
$ ls easy_rev_simple_encryptor/rev_simpleencryptor
encrypt flag.enc
(kali@barengsinau)~[~/lab/ailab/HTB_Challenge]
$
```

Web Browser (HTB Labs):

The browser shows the HTB Labs interface with a sidebar containing navigation links: Home, Profile, Friends, Unranked Season 9 Tier, Rank, Season 9, Starting Point, Machines, Sherlocks, Challenges (selected), Tracks, Pro Labs, Fortress, Rankings, and Universities.

The main content area displays the "Simple Encryptor" challenge page. It includes a "Play Challenge" button and a description: "On our regular checkups of our secret flag storage server we found out that we were hit by ransomware, but alas the encryptor program didn't...". Below the description, there is a section for "Simple Encryptor" with a "Flag" field and a "Submit" button.

Terminal Output (Right Panel):

```
2025-12-14 01:33:02,187 [INFO] Initialized Kali Tools Client connecting to http://localhost:5000
2025-12-14 01:33:02,189 [ERROR] Request failed: HTTPConnectionPool(host='localhost', port=5000): Max retri
url: /health (Caused by NewConnectionError('<urllib3.connection.HTTPConnection object at 0x7fa70de66900>:
sh a new connection: [Errno 111] Connection refused'))
2025-12-14 01:33:02,190 [WARNING] Unable to connect to Kali API server at http://localhost:5000: Request f
tionPool(host='localhost', port=5000): Max retries exceeded with url: /health (Caused by NewConnectionErro
ction.HTTPConnection object at 0x7fa70de66900): Failed to establish a new connection: [Errno 111] Connecti
2025-12-14 01:33:02,190 [WARNING] MCP server will start, but tool execution may fail
2025-12-14 01:33:02,206 [INFO] Starting Kali MCP server
^C^Z
zsh: suspended python3 mcp_server.py

(kali@barengsinau)~[~/tools/mcp/MCP-Kali-Server]
$ python3 kali_server.py
2025-12-14 01:46:52,754 [INFO] Starting Kali Linux Tools API Server on port 5000
* Serving Flask app 'kali_server'
* Debug mode: off
2025-12-14 01:46:52,759 [INFO] WARNING: This is a development server. Do not use it in a production deploy
ment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://192.168.0.108:5000
2025-12-14 01:46:52,759 [INFO] Press CTRL+C to quit
2025-12-14 01:47:18,931 [INFO] Executing command: which nmap
2025-12-14 01:47:18,938 [INFO] Executing command: which gobuster
2025-12-14 01:47:18,942 [INFO] Executing command: which dirb
2025-12-14 01:47:18,946 [INFO] Executing command: which nikto
2025-12-14 01:47:18,949 [INFO] 127.0.0.1 - - [14/Dec/2025 01:47:18] "GET /health HTTP/1.1" 200 -
2025-12-14 01:47:32,769 [INFO] Executing command: which nmap
2025-12-14 01:47:32,775 [INFO] Executing command: which gobuster
2025-12-14 01:47:32,780 [INFO] Executing command: which dirb
2025-12-14 01:47:32,790 [INFO] Executing command: which nikto
2025-12-14 01:47:32,797 [INFO] 127.0.0.1 - - [14/Dec/2025 01:47:32] "GET /health HTTP/1.1" 200 -
[] The original flag data is now gone to be found, but luckily we not only have the encryptor

zsh: corrupt history file /home/kali/.zsh_history
(kali@barengsinau)~[~/tools/mcp/MCP-Kali-Server]
$ python3 mcp_server.py
2025-12-14 01:47:18,923 [INFO] Initialized Kali Tools Client connecting to http://localhost:5000
2025-12-14 01:47:18,950 [INFO] Successfully connected to Kali API server at http://localhost:5000
2025-12-14 01:47:18,950 [INFO] Server health status: healthy
2025-12-14 01:47:18,965 [INFO] Starting Kali MCP server
[]
```



Agentic AI 101

Make a machine “Work”



Era 5 – Agentic AI (2023-Now)

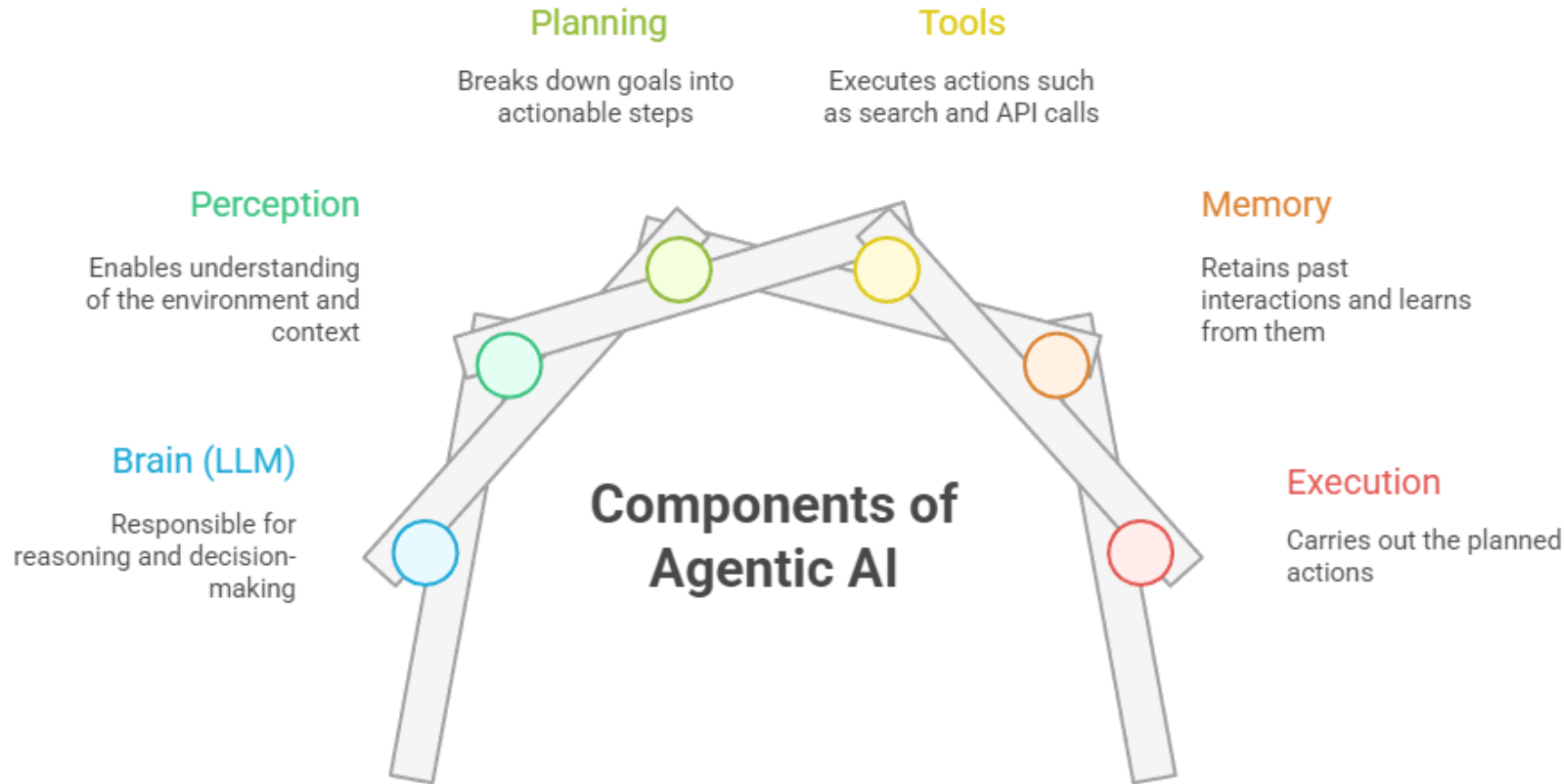
Definition:

- AI system yang bisa secara AUTONOMOUS:
 - Merencanakan langkah-langkah
 - Menggunakan tools
 - Bertindak di environment
 - Mencapai goals tanpa hand-holding

Perbedaan Utama:

- LLM Traditional → Responsive (Q&A)
- Agentic AI → Proactive (Goal-oriented)

Anatomy of an AI Agent



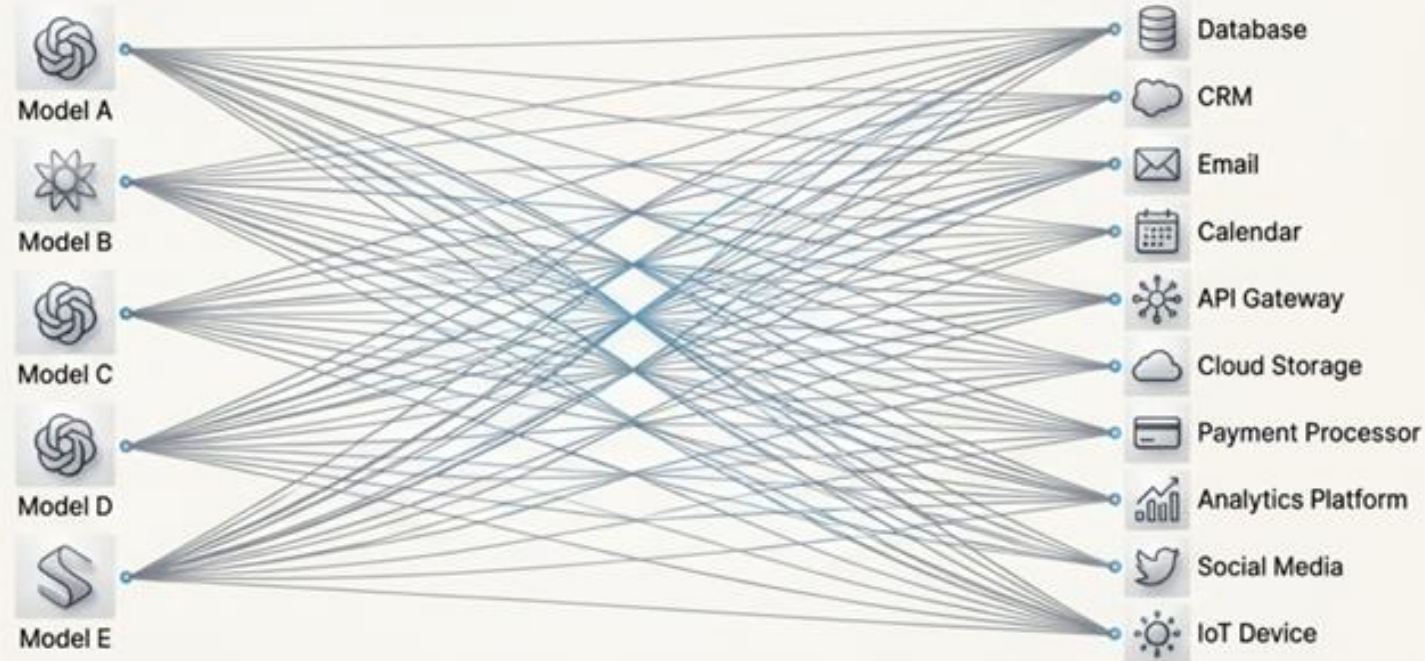
From Static Model to Orchestrated System

Feature	Traditional LLM	Agentic AI
Primary Goal	Content generation and conversational response	Goal setting, multi-step problem solving, autonomous action
Knowlede Base	Parametric memory (frozen at training) or single-source RAG	Hybrid memory (RAG + experience) and real-time external data access
Interaction	Passive (Text input/output)	Active (Tool execution, external system interaction via protocols)
Complexity handling	Limited by context window	Solves complex, multi-step problem via planning and decomposition
Key limitation	Factual accuracy and up-to-date knowledge	Inability to act and lack of integrated memory/learning

Apa itu MCP?



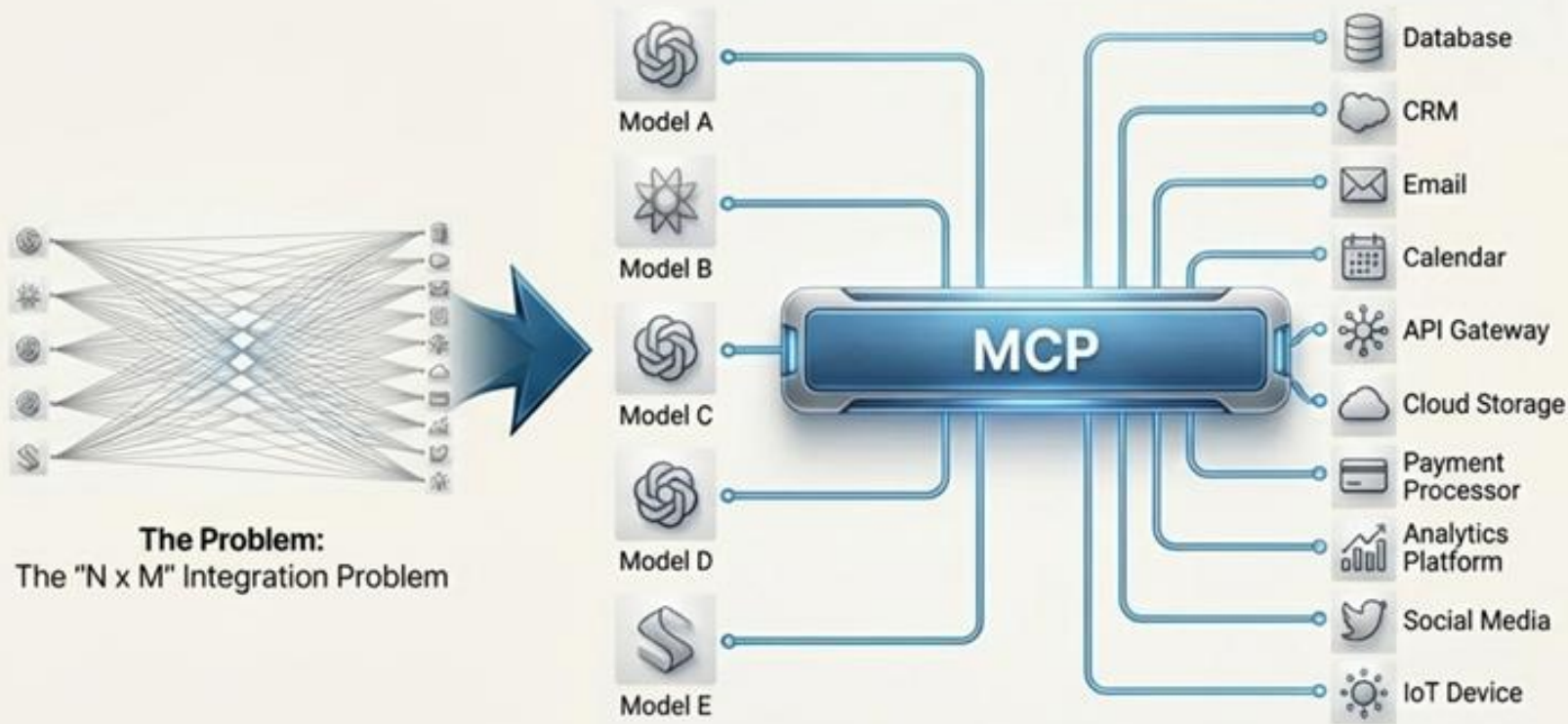
How Agent “Actually” Work?



The “N x M” Integration Problem

This approach creates a fragile and unscalable ecosystem. For every N AI models and M external tools, a unique connector is needed, leading to an explosion in development and maintenance complexity.

The Universal Interface: The Model Context Protocol (MCP)



The Problem:
The "N x M" Integration Problem

**MCP is the USB-C port
for AI applications.**

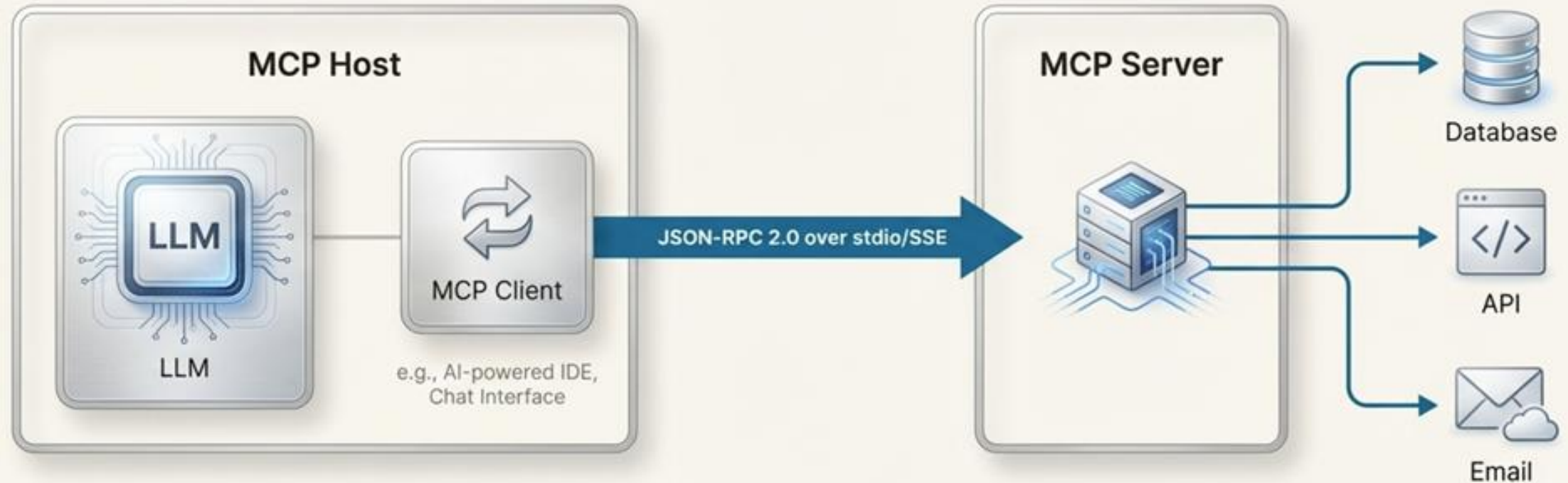
Definition

MCP is an open protocol that provides a standardized "language" for LLMs to communicate with external data, applications, and services. It formalizes and builds upon existing concepts like tool use and function calling.

Key Function

It moves agents beyond simple chat and into the realm of intelligent automation by enabling secure connections to enterprise systems.

The MCP Architecture: Host, Client, and Server



MCP Host

The AI application containing the LLM.
The user's interaction point.

MCP Client

Located within the Host, it translates the LLM's planned action into the MCP format and translates the server's observation back for the LLM's reflection phase.

MCP Server

The external system that exposes a data source or capability (e.g., `'database_query'`, `'email_sender'`) via the protocol.

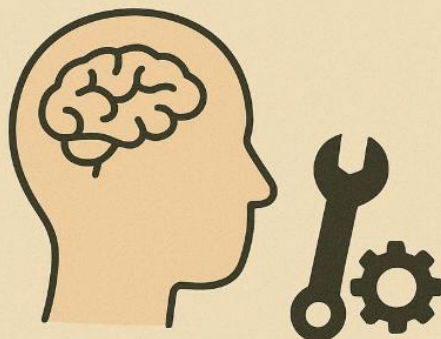
tl:dr

LLM ALONE



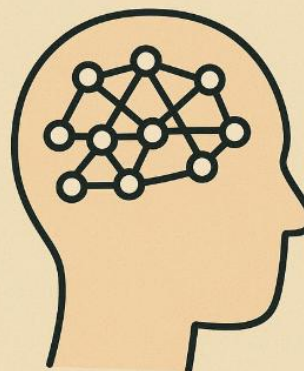
**SMART
BUT BLIND**

LLM + TOOLS

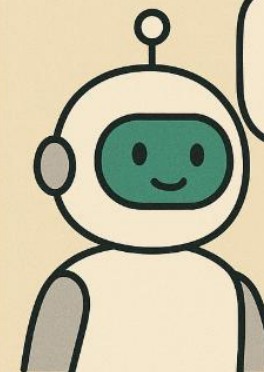


**SMART
+ CAPABLE**

TRADITIONAL AI

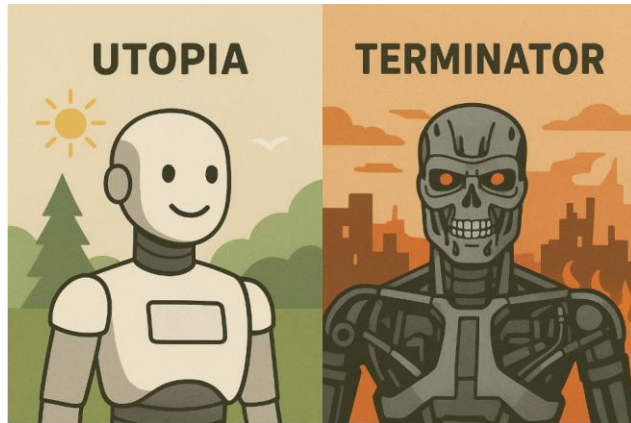


AGENTIC AI

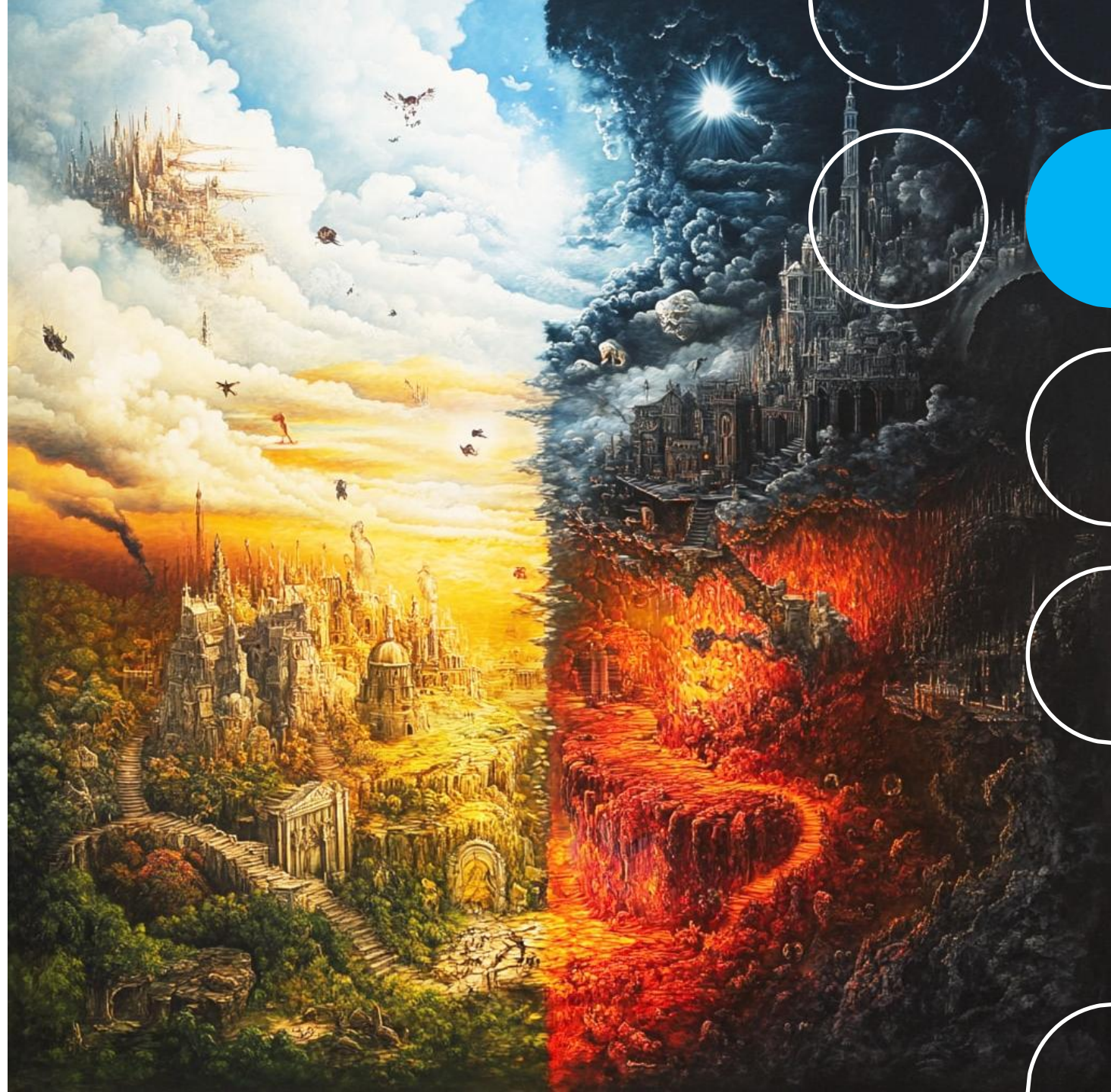


I can
help!

Future: Utopia / Dystopia (?)



<https://gregoreite.com/fear-not-ai-utopia-ai-dystopia-all-at-once-embracing-the-contradiction/>



DEMO & AMA



ASK ME ANYTHING

